

巴适声工厂隐私版使用手册 / Bashi Voice Factory Privacy Edition User Guide

Bashi Voice Factory Privacy Edition (巴适声工厂 · 隐私版)

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Bashi Voice Factory Privacy Edition (巴适声工厂 · 隐私版)

License MIT version 0.1.1 python 3.12 embed platform Windows

Version: 0.1.1

A fully offline desktop web app for high-quality text-to-speech and speech-to-text. After the one-time first-launch download, **everything runs on your own machine** — TTS synthesis, audio export, transcription, and storage. No audio data ever leaves your computer.

⚡ Need cloud-quality voices and faster generation? Check out [Bashi Voice Factory Turbo \(巴适声工厂 · 极速版\)](#) — Microsoft Edge TTS, 14 languages, 50,000-character long text. Requires internet.

Author: Alex Li (ncorecpu@gmail.com) **License:** [MIT License](#) **Source code:** <https://github.com/gtree965/bashi-voice-factory-privacy> **Download:** [GitHub Releases](#) • [files.fm mirror](#)

✨ Highlight Features

🔒 Fully Local Text-to-Speech

- **Local Qwen3-TTS-12Hz-1.7B-CustomVoice** running entirely on your machine — no cloud calls during synthesis.
- **9 curated voice presets** with native-language sample previews built in.
- **10 languages** supported: Chinese (Mandarin, Cantonese, Sichuanese, Beijing, Northeast), English (US/UK), Japanese, Korean, German, French, Russian, Portuguese, Spanish, Italian.
- **Instruction control** for style, accent, emotion via natural-language prompts.

🎯 Automatic Backend Selection

- **GGUF + Vulkan** for AMD / Intel / iGPU users (default, fast, RAM-friendly).
- **GGUF + CUDA** for NVIDIA users via a one-click in-app upgrade (optional ~595 MB add-on download; no manual weight setup). New in v0.1.1.
- **CPU fallback** for entry-level hardware (works on Intel N100-class boxes).
- One-click **speed test** calibrates ETA estimates to your specific machine.

🎤 Local Offline Speech-to-Text (STT)

- **SenseVoice Small** (default, 242 MB) — multilingual zh/en/ja/ko/yue, non-autoregressive (fast on CPU).

- **Parakeet TDT 0.6B** (optional, 661 MB) — English specialist, NVIDIA's ~1.7% WER champion.
- **VAD-based segmentation** via Silero — accurate timestamps, no chunk-boundary stutter.
- **Live transcript streaming** via SSE; exports to TXT / SRT / VTT.

Smart First-Launch Download with Resume

- **HTTP Range / resume** built in — a transient WiFi drop won't restart your 2.2 GB GGUF download from zero.
- **3-attempt retry** with exponential backoff for both pip dependencies and model files.
- **China-friendly mirrors:** Aliyun PyPI mirror auto-detected by locale; GGUF runtime hosted on ModelScope; STT models hosted on hf-mirror.cn.

LAN Sharing for Phone / Tablet Use

- First-launch prompt: bind to 0.0.0.0 to allow access from any device on the same WiFi.
 - Compatible with mobile browsers — type the LAN IP and use it like a local web app.
-

Quick Start

Easiest path

1. Unzip the downloaded file to any folder (Desktop or wherever you like).
2. **Double-click** `start_启动.bat` at the top level of the extracted folder.
3. First launch will:
 - Install Python dependencies (~700 MB, 8-15 min depending on network/CPU)
 - Automatically download the GGUF model when missing (~2.2 GB, 2-15 min depending on network; Ctrl+C cancels and the next launch resumes)
 - Open `http://127.0.0.1:5050` in your default browser
4. After first launch, **all subsequent launches are offline** and start in ~5 seconds.

Advanced

Users who prefer to bypass the top-level launcher can run `bashi-privacy-app\run_portable.bat` directly. Both paths are identical.

Network Behavior

Privacy commitment: this app does not phone home, does not upload audio, does not collect telemetry. **The app never automatically checks any internet resource.** Every network operation is documented below and requires explicit user action — no background polling, no analytics, no silent update probes.

First launch only (one-time, ~700 MB + ~2.2 GB):

- pip dependencies from <https://mirrors.aliyun.com/pypi/simple/> (or PyPI default outside China)
- GGUF model from <https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime>
- STT model (when user opts in) from <https://hf-mirror.com/csukuangfj/...> (or HuggingFace fallback)

Runtime network activity: zero. TTS synthesis, STT transcription, and audio export are 100% on-device.

Optional CUDA acceleration (NVIDIA users, user-initiated only): when an NVIDIA GPU is detected, the UI shows an "Upgrade to CUDA acceleration" banner. Clicking it downloads a one-time ~595 MB CUDA runtime add-on from <https://modelscope.cn/models/gtree592/bashi-qwen3-tts-cuda-runtime>. This is entirely optional — the Vulkan path works on NVIDIA without it. Nothing downloads unless you click.

Manual update check (user-initiated only, never automatic): the UI footer "Check for updates" button opens a release page in a new browser tab. You can also check manually anytime at:

- GitHub Releases: <https://github.com/gtree965/bashi-voice-factory-privacy/releases>
- files.fm mirror: <https://files.fm/u/juvstxmrez>

Air-gapped / sensitive deployment (banks, healthcare, government, internal corporate networks): after first-launch downloads complete on a connected machine, the entire bashi-voice-factory-privacy-v0.x.0/ extracted folder can be copied to an offline / physically-isolated machine and run with zero further network access required. The app has no telemetry endpoints to firewall, no auto-update worker, no callback URLs. To verify, monitor outbound traffic with Wireshark / Resource Monitor during a normal session — it should be silent.



Hardware Expectations

The table below shows **actually measured** numbers from author hardware. The built-in speed test (click 测速 / Speed Test on the right-hand panel) calibrates the ETA panel to your specific machine on first use, so you don't have to extrapolate from this table.

Hardware	Auto-selected backend	"你好。" probe (25 chars)	1,000-char synthesis ETA	Status
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	~3 s	a few minutes	✓ Tested 2026-05
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 s	~5-10 min	✓ Tested
Intel N305 laptop + UHD iGPU	GGUF + Vulkan / DirectML	53 s	25-46 min	✓ Tested 2026-05-25
Intel N100 mini-PC + UHD iGPU	GGUF + Vulkan / DirectML	126 s	58 min - 1h49 min	✓ Tested 2026-05-25
NVIDIA RTX / GTX (desktop)	GGUF + CUDA (one-click in-app upgrade)	community reports welcome	community reports welcome	awaiting desktop tester reports via GitHub Issues

Hardware	Auto-selected backend	"你好。" probe (25 chars)	1,000-char synthesis ETA	Status
Apple Silicon / Intel Arc	—	not yet measured	not yet measured	not validated yet

⚠️ **Entry-level CPUs (Intel N100 / N305 class) are only practical for short text — under ~200 characters per generation.** A 5,000-character essay would take 2-9 hours on these boxes. For long-form audio (lectures, audiobooks), please use discrete-GPU hardware (AMD RX 500/600/9000 series, NVIDIA RTX class).

ℹ️ **NVIDIA users (desktop RTX / GTX):** the default path is GGUF + Vulkan (NVIDIA cards support Vulkan via the proprietary driver). For native CUDA acceleration, click the one-click in-app upgrade banner (v0.1.1+) — it downloads a ~595 MB CUDA runtime add-on, no manual weight setup required. **Requires NVIDIA driver ≥ 545.x for the CUDA 12.4 runtime.** Measured token/sec numbers from desktop RTX/GTX testers welcome via [GitHub Issues](#). Cloud datacenter NVIDIA cards (A10/A100/T4) running in TCC mode have additional setup steps — see Troubleshooting below.

🧩 **Hardware Coverage Matrix**

Beyond the specific machines benchmarked above, here is how the auto-selected acceleration path maps across mainstream Windows hardware:

Hardware class	Acceleration path used	Status
NVIDIA RTX 30 / 40 / 50 (desktop)	GGUF + CUDA (optional in-app upgrade) • Vulkan default	✅ Build verified (download + extract + DLL load); desktop tester measured speeds welcome via Issues. CUDA add-on requires driver ≥ 545.x.
NVIDIA GTX 10 / 16 (desktop)	GGUF + CUDA (optional in-app upgrade) • Vulkan default	✅ Same flow, same driver requirement.

Hardware class	Acceleration path used	Status
NVIDIA datacenter (A10 / A100 / T4)	Manual setup required	⚠️ TCC mode + old cloud driver = manual workaround. See Troubleshooting.
AMD RX 500 / 600 / 7000 / 9000 (discrete)	GGUF + Vulkan + DirectML	✅ Tested (RX 590, RX 9060 XT)
Intel Arc A-series (A380 / A580 / A750 / A770)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
Intel iGPU (UHD / Iris Xe / Arc iGPU)	GGUF + Vulkan + DirectML	✅ Tested (Intel N305 UHD)
AMD APU iGPU (Vega 7/8, RDNA 2/3)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
CPU only (no usable GPU / no driver)	GGUF + CPU (ggml-cpu auto-SIMD)	✅ Works; auto-selected only when no GPU detected
NPU (Snapdragon X / Lunar Lake / Ryzen AI 300)	—	❌ Not yet utilized
ARM64 Windows (Snapdragon X Copilot+ PC)	—	❌ Not supported (runs via slow x64 emulation)

⚙️ Known Limitations (roadmap)

- **NVIDIA CUDA is opt-in, not bundled.** By default NVIDIA cards use the Vulkan path; native CUDA acceleration requires the one-click in-app upgrade (~595 MB add-on, v0.1.1+). This keeps the main download small for the AMD / Intel / CPU majority. Requires NVIDIA driver $\geq 545.x$.
- **Weak iGPU may be slower than pure CPU.** On Intel N100-class hardware, DirectML / Vulkan transfer + scheduling overhead can outweigh GPU benefit. A/B test by launching from a cmd window after `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` — feedback on real speed numbers welcome.

- **NPU acceleration is not yet used** (Snapdragon X Elite, Intel Lunar Lake AI Boost, AMD Ryzen AI 300). Under investigation for v0.2+.
- **ARM64 Windows is not natively supported** (Surface Pro 11, Galaxy Book4 Edge, etc.). x64 emulation works but is slow. Native ARM64 build is a v0.3+ candidate.



Troubleshooting

Symptom	Likely cause / fix
Start_启动.bat shows "Array index expression is missing"	Old zip — re-download the latest from GitHub Releases or the files.fm mirror; the BOM bug was fixed in v0.1.0 final
pip install stops mid-way after WiFi blip	Retry triggers automatically (3 attempts, 5s/30s/120s backoff). If all 3 fail, fix network and re-run the launcher — pip caches what's already installed
pip install fails with a "long path support" message	Run the registry one-liner from the launcher's advisory in an Administrator PowerShell, then re-launch
GGUF download interrupted	Same: retries with HTTP Range/resume; re-run launcher to continue from .part file
App exits with "No usable backend was found"	Check launch_log.txt — usually GGUF runtime DLL missing, GPU driver outdated, or RAM <8 GB. App now prints a bilingual advisory with specific causes
STT download says "镜像失败"	Should not happen in v0.1.0 final — old behavior from removed ModelScope path
Cloud / datacenter NVIDIA GPU (A10 / A100 / T4 on Chinese cloud Windows images): access violation reading 0x0000000000000000 OR GGUF probe failed	Datacenter NVIDIA GPUs typically run in TCC mode (Vulkan/DirectML disabled) with older drivers (~538.x) incompatible with the CUDA 12.4 add-on. Desktop RTX/GTX cards are NOT affected (they run WDDM mode by default with modern drivers). Cloud workaround for v0.1.1: (1) rename vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll to .disabled to avoid Vulkan ICD crash; (2) edit bashi-privacy-app\run_portable.ps1 and add \$env:USE_GGUF_BACKEND = "1" + \$env:GGUF_ONNX_PROVIDER

Symptom	Likely cause / fix
	= "CPU" after the Remove-Item Env:USE_GGUF_BACKEND lines (~line 270) to bypass the probe-ladder env scrub; (3) pre-install CUDA add-on via CLI python download_cuda_runtime.py. Requires NVIDIA driver ≥ 545.x for CUDA 12.4 — older cloud drivers won't work and you'll see CUDA init crash. v0.1.2 will auto-detect this scenario and apply the workaround.

Full log path: bashi-privacy-app\launch_log.txt

What's in the Zip

bashi-voice-factory-privacy-v0.1.1/	
├─ Start_启动.bat	← double-click here
├─ Start_CPU_only_仅CPU启动.bat	← force CPU mode
(entry-level iGPU A/B)	
├─ README.md	← this file
├─ README_CN.md	← Chinese version
├─ LICENSE	← MIT license
├─ VERSION	← release version
├─ 巴适声工厂隐私版使用手册	
_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf	
	← help PDF
(bilingual)	
├─ bashi-privacy-app/	← app code +
embedded Python	
└─ run_portable.bat	← same launcher,
direct path	
└─ README.md	← project README
copy	
└─ ...	
└─ vulkan_backend_spike/	← GGUF runtime
(populated on first launch)	
└─ Qwen3-TTS-GGUF/	

License

MIT License © 2026 Alex Li

Bundled third-party components retain their original licenses:

- **Qwen3-TTS-12Hz-1.7B-CustomVoice:** Tongyi Lab (Alibaba), Apache 2.0

- **GGUF runtime:** based on HaujetZhao/Qwen3-TTS-GGUF
 - **SenseVoice Small / Silero VAD / Parakeet TDT:** see respective model cards
-

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Issues, feedback, and feature requests welcome via the GitHub release page or email.

[English](#) | 中文文档

巴适声工厂 · 隐私版 (Bashi Voice Factory Privacy Edition)

License MIT version 0.1.1 python 3.12 embed platform Windows

版本: 0.1.1

完全离线运行的本地语音工厂网页应用。首次启动联网下载完依赖与模型之后，所有文字转语音、语音转文字、音频生成、文件保存都在你自己的电脑上完成，没有任何音频数据上传云端。

⚡ 想要云端语音质量、更快的合成速度？请关注 [巴适声工厂 · 极速版 \(Bashi Voice Factory Turbo\)](#) — 微软 Edge TTS 引擎、14 种语言、5 万字长文。需要联网。

作者: Alex Li (ncorecpu@gmail.com) 许可: [MIT License](#) 源码: <https://github.com/gtree965/bashi-voice-factory-privacy> 下载: [GitHub Releases](#) · [files.fm](#) 镜像

✨ 核心亮点

🔒 完全本地文字转语音

- **Qwen3-TTS-12Hz-1.7B-CustomVoice 本地推理** — 合成过程零云端调用。
- **9 个精选音色**，内置母语试听样例。
- **10 种语言**：中文（普通话、粤语、四川话、北京话、东北话）、英文（美/英）、日语、韩语、德语、法语、俄语、葡萄牙语、西班牙语、意大利语。
- **自然语言指令控制** — 风格、口音、情绪都可通过 instruct 提示调整。

🎯 后端自动选择

- **GGUF + Vulkan**：AMD / Intel / 集成显卡用户默认路线（速度快、内存友好）。
- **GGUF + CUDA**：NVIDIA 用户一键应用内升级（可选 ~595 MB 附加包下载，无需手动配置权重）。v0.1.1 新增。

- **CPU 回退**：入门级硬件（如 Intel N100 系列）也能跑。
- 一键 **测速** 校准 ETA 估算，让等待时间贴合你这台机器。

本地离线音视频转文字 (STT)

- **SenseVoice Small**（默认，242 MB）— 中英日韩粤多语种，非自回归架构，CPU 上速度快。
- **Parakeet TDT 0.6B**（可选，661 MB）— 英文专用，NVIDIA 出品，WER 约 1.7%。
- **Silero VAD 精准分段** — 时间戳准确、零重叠、零卡顿。
- **实时滚屏字幕 SSE 流式输出**；支持导出 TXT、SRT、VTT。

智能首次下载，断点续传

- **HTTP Range / resume 内置** — 一次 WiFi 闪断不会让 2.2 GB 的 GGUF 下载从零开始。
- **3 次重试，指数退避** — pip 依赖与模型文件都有同样的兜底逻辑。
- **国内友好镜像**：自动识别中国时区使用阿里云 PyPI；GGUF 运行模型走 ModelScope；STT 模型走 hf-mirror.cn。

局域网共享，手机平板可访问

- 首次启动询问是否绑定 0.0.0.0，允许同一 WiFi 下其他设备访问。
- 手机浏览器输入电脑的局域网 IP 即可使用，跟本地网页一样流畅。

快速上手

最简流程

1. 把下载的 zip 解压到任意文件夹（桌面或随便哪里都行）。
2. 在解压后的顶层目录里 **双击** `Start_启动.bat`。
3. 首次启动会：
 - 安装 Python 依赖（约 700 MB，根据网速 / CPU 不同需要 8-15 分钟）
 - 缺少 GGUF 模型时自动下载（约 2.2 GB，2-15 分钟；可按 Ctrl+C 取消，下次启动会续传）
 - 自动在默认浏览器打开 `http://127.0.0.1:5050`
4. 首次完成后，**之后每次启动都是离线**，5 秒左右即可打开。

进阶

如果想跳过顶层启动器，可以直接运行 `bashi-privacy-app\run_portable.bat`，效果完全一样。

网络行为说明

隐私承诺：本程序不主动联网、不上传音频、不收集使用数据。**程序永远不会自动检查任何在线资源。**所有联网行为均在下文有完整说明，且需要用户主动触发 — 无后台轮询、无统计分析、无静默更新探测。

仅首次启动（一次性，约 700 MB + 约 2.2 GB）：

- pip 依赖来自 <https://mirrors.aliyun.com/pypi/simple/>（海外用户默认 PyPI）
- GGUF 模型来自 <https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime>
- STT 模型（用户主动点击下载）来自 <https://hf-mirror.com/csukuangfj/...>（备用 HuggingFace）

运行时网络行为：零。文字转语音、音视频转文字、音频导出全部本机完成。

可选 CUDA 加速（NVIDIA 用户，仅用户主动触发）：检测到 NVIDIA 显卡时，UI 会显示"升级到 CUDA 加速"横幅。点击后从

<https://modelscope.cn/models/gtree592/bashi-qwen3-tts-cuda-runtime> 下载一次性 ~595 MB CUDA 运行时附加包。完全可选 — 不下载也能用 Vulkan 路线。不点就不会下载任何东西。

手动更新检查（仅用户主动触发，永不自动）：UI 右下角"查看最新版本"按钮在新浏览器标签页打开发布页。也可以随时手动访问：

- GitHub Releases: <https://github.com/gtree965/bashi-voice-factory-privacy/releases>
- files.fm 镜像: <https://files.fm/u/juvstxmrez>

离网 / 敏感场景部署（银行、医疗、政务、企业内网）：在联网机器上完成首次启动的所有下载后，整个 `bashi-voice-factory-privacy-v0.x.0/` 解压文件夹可以原样拷贝到离网/物理隔离机器上运行，之后所有功能均无需任何网络访问。程序没有任何遥测端点需要防火墙拦截，没有自动更新守护进程，也没有回调 URL。可用 Wireshark / 资源监视器在正常使用过程中监测出站流量验证 — 应当全程静默。

 硬件要求

下表是**作者实测**的数据。首次运行后点右侧面板的"测速"按钮，程序会根据你这台机器实际表现重新校准 ETA 估算 — 不用照搬此表。

硬件	自动选择的后端	"你好。"探测 (25 字)	1,000 字合成预计	状态
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	约 3 秒	几分钟	✓ 2026-05 实测
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 秒	约 5-10 分钟	✓ 实测
Intel N305 笔记本 + UHD 集显	GGUF + Vulkan / DirectML	53 秒	25-46 分钟	✓ 2026-05-25 实测
Intel N100 迷你主机 + UHD 集显	GGUF + Vulkan / DirectML	126 秒	58 分钟 - 1 小时 49 分	✓ 2026-05-25 实测
NVIDIA RTX / GTX (桌面)	GGUF + CUDA (一键应用内升级)	待社区报数	待社区报数	等待桌面 tester 通过 GitHub Issues 报数
Apple Silicon / Intel Arc	—	暂未实测	暂未实测	尚未验证

 **入门级 CPU (Intel N100 / N305 一类) 仅适合短句试用 — 单次生成建议不超过 200 字。** 这类机器跑 5,000 字长文需要 2-9 小时。如果想做长文音频 (讲座、有声书)，请用独立显卡硬件 (AMD RX 500/600/9000 系、NVIDIA RTX 类)。

 **NVIDIA 用户 (桌面 RTX / GTX)：**默认路线是 GGUF + Vulkan (NVIDIA 驱动支持 Vulkan)。想要原生 CUDA 加速，点击应用内一键升级横幅 (v0.1.1+) — 会下载 ~595 MB CUDA 运行时附加包，无需手动配置权重。**需要 NVIDIA 驱动 ≥ 545.x 才能使用 CUDA 12.4 运行时。** 桌面 RTX/GTX 用户的实测 token/sec 数字欢迎通过 [GitHub Issues](#) 反馈。云端数据中心 NVIDIA 显卡 (A10/A100/T4) 在 TCC 模式下需要额外设置 — 见下方常见问题排查。

 硬件兼容性矩阵

除了上表里具体测过的几台机器，下表说明主流 Windows 硬件在自动后端选择下走的加速路径：

硬件类别	实际加速路径	状态
NVIDIA RTX 30 / 40 / 50（桌面）	GGUF + CUDA （可选应用内升级）• 默认 Vulkan	✅ 构建已验证（下载 + 解压 + DLL 加载流程）；欢迎桌面 tester 通过 Issues 报实测速度。CUDA add-on 需驱动 ≥ 545.x。
NVIDIA GTX 10 / 16（桌面）	GGUF + CUDA （可选应用内升级）• 默认 Vulkan	✅ 同一流程，同样驱动要求。
NVIDIA 数据中心卡（A10 / A100 / T4）	需要手动配置	⚠️ TCC 模式 + 云端老驱动 = 需要手动 workaround。见常见问题排查。
AMD RX 500 / 600 / 7000 / 9000（独立显卡）	GGUF + Vulkan + DirectML	✅ 实测（RX 590、RX 9060 XT）
Intel Arc A 系列（A380 / A580 / A750 / A770）	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
Intel 集显（UHD / Iris Xe / Arc iGPU）	GGUF + Vulkan + DirectML	✅ 实测（Intel N305 UHD）
AMD APU 集显（Vega 7/8、RDNA 2/3）	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
纯 CPU（无可用 GPU / 无驱动）	GGUF + CPU （ggml-cpu 自动 SIMD）	✅ 可用；仅在无 GPU 时自动选择
NPU（Snapdragon X / Lunar Lake / Ryzen AI 300）	—	❌ 暂未利用
ARM64 Windows（Snapdragon X Copilot+ PC）	—	❌ 暂不支持（走 x64 仿真较慢）

⚙️ 已知限制（路线图）

- **NVIDIA CUDA 为可选项，不内置。** NVIDIA 显卡默认走 Vulkan 路线；原生 CUDA 加速需点击应用内一键升级（~595 MB 附加包，v0.1.1+）。这样主下载包对 AMD / Intel / CPU 大多数用户保持精简。需 NVIDIA 驱动 ≥ 545.x。
- **弱集显可能比纯 CPU 还慢。** Intel N100 类硬件上，DirectML / Vulkan 的数据传输与调度开销可能盖过 GPU 收益。可在 cmd 窗口里 `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` 后再启动来 A/B 测试，欢迎反馈实测数字。
- **NPU 加速尚未利用**（Snapdragon X Elite、Intel Lunar Lake AI Boost、AMD Ryzen AI 300）。v0.2+ 调研中。
- **ARM64 Windows 暂未原生支持**（Surface Pro 11、Galaxy Book4 Edge 等）。x64 仿真能跑但速度受限。原生 ARM64 构建是 v0.3+ 候选。

🐛 常见问题排查

现象	可能原因 / 处理
双击 Start_启动.bat 报 "Array index expression is missing"	旧版 zip。请从 GitHub Releases 或 files.fm 镜像重新下载最新版；该 BOM 问题在 v0.1.0 正式版已修复
pip 安装途中 WiFi 闪断后中止	自动重试 3 次（5/30/120 秒退避）。3 次都失败时，修好网络后重新运行启动器即可（pip 会跳过已装好的包）
pip 安装失败并提示 "long path support"	按启动器提示，在管理员 PowerShell 中运行一次注册表命令，然后重新启动
GGUF 下载中断	同样自动重试，且支持 HTTP Range 续传，重新运行启动器从 .part 文件续传
启动报 "No usable backend was found"	查 launch_log.txt。常见原因：GGUF 运行 DLL 缺失、显卡驱动过旧、可用内存 <8 GB。程序会打印中英双语提示告知具体原因
STT 下载闪过 "镜像失败"	v0.1.0 正式版不会出现 — 是旧版残留的 ModelScope 路径，已删除
云端 / 数据中心 NVIDIA 显卡（A10 / A100 / T4，国内云 Windows 镜像）：access violation	数据中心 NVIDIA 显卡通常运行在 TCC 模式 （Vulkan/DirectML 被屏蔽），且驱动版本较旧（~538.x）跟 CUDA 12.4 add-on 不兼容。 桌面 RTX/GTX 显卡不受影响 （默认 WDDM 模式 + 现代驱动）。v0.1.1 云端 workaround：(1) 把

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**: 通义实验室（阿里巴巴）Apache 2.0
 - **GGUF runtime**: 基于 HaujetZhao/Qwen3-TTS-GGUF
 - **SenseVoice Small / Silero VAD / Parakeet TDT**: 详见各自模型卡
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欢迎通过 [GitHub release](#) 页面或邮件提交 issue、反馈或功能建议。